THE FIBRE OF A FIBRED CATEGORY IN GROUPOIDS IS EQUIVALENT AS A SITE TO THE BASE SLICE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Fibred in groupoids definition. categories-definition-fibred-groupoids (cross-chapter, named). Gives that x'/x maps to p(x')/p(x) is an equivalence of categories.
- Inherited topology definition. definition-topology-inherited (same chapter, named). Identifies coverings of x' in X/x with coverings of p(x') in C/U, giving the site isomorphism.

Source reference. Original reference 003T: focused-seeds/geometry/S9-stk-localizing/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 06NV: sources/stacks-stacks/pinned-original-source.tex, source lines 2255–2255. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $p : \mathcal{X} \to \mathcal{C}$ be a category fibred in groupoids. Let $x \in \mathrm{Ob}(\mathcal{X})$ lying over $U = p(x)$. The functor p induces an equivalence of sites $\mathcal{X}/x \to \mathcal{C}/U$ where $\mathcal{X}$ is endowed with the topology inherited from $\mathcal{C}$.*

Proof. Here $\mathcal{C}/U$ is the localization of the site $\mathcal{C}$ at the object U and similarly for $\mathcal{X}/x$. It follows from Categories, Definition 003T that the rule $x'/x \mapsto p(x')/p(x)$ defines an equivalence of categories $\mathcal{X}/x \to \mathcal{C}/U$. Whereupon it follows from Definition 06NV that coverings of x' in $\mathcal{X}/x$ are in bijective correspondence with coverings of $p(x')$ in $\mathcal{C}/U$. □

Date: stacks-project:stacks.tex:local-snapshot-20261012; GNU FDL 1.2 or later.